BUS 9430 · CAPSTONE 2 · DATA UNDERSTANDING AND PREPARATION

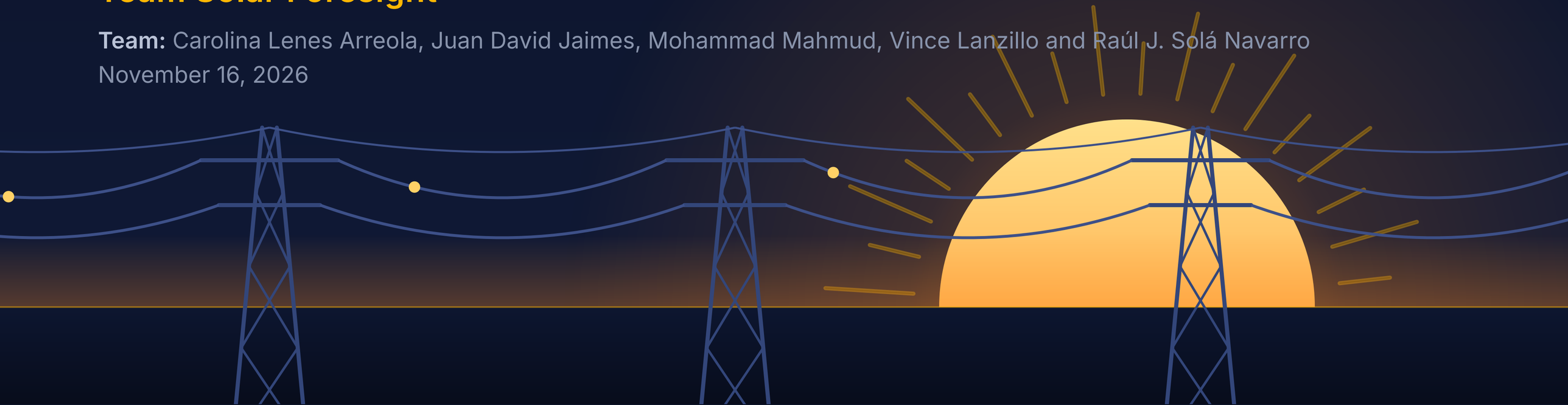
Tomorrow's Sun, Today's Plan

What our data shows, and how we get it ready

Team Solar Foresight

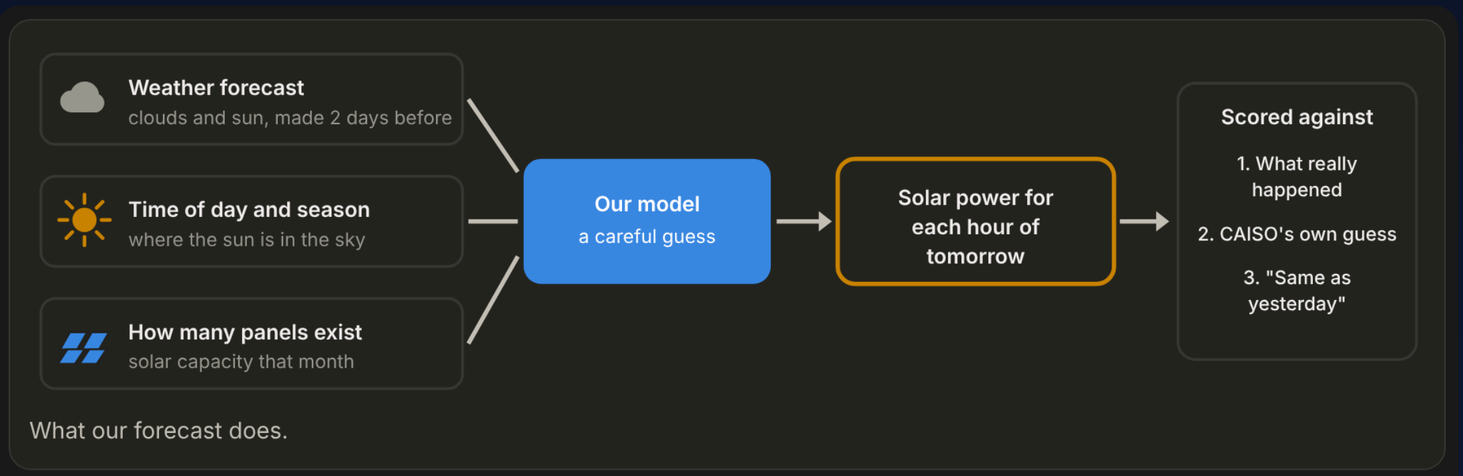
Team: Carolina Lenes Arreola, Juan David Jaimes, Mohammad Mahmud, Vince Lanzillo and Raúl J. Solá Navarro

November 16, 2026



THE BUSINESS PROBLEM

Guessing tomorrow’s solar power, hour by hour



What our forecast does.

Why it matters

- Fewer last-minute power buys.
- Less power wasted.
- Fewer backup gas plants running.

Who it helps

- Grid planners.
- Companies that buy power.
- Anyone who pays an electric bill.

How it is different

- Only free public data.
- A fair scorecard against CAISO and simple guesses.
- Every step shown, so anyone can check it.

TARGET VARIABLE

Solar power for each hour of tomorrow

In megawatts (MW), for the whole CAISO grid.

THE RULE

Only what we know at 10 a.m.

CAISO's bids close at 10 a.m. Pacific, the day before.

Picture from our plain-language primer (docs/solar-primer.html in the team repository).

OBTAIN AND QUALIFY THE DATA

Five public sources, each picked for a reason

SOURCE	WHAT IT GIVES US	WHY WE CHOSE IT
EIA hourly grid data	Actual solar output, hour by hour. Our answer key.	Official U.S. data for the whole CAISO area.
Open-Meteo	Weather forecasts made 2 days before, at 51 spots in California.	Keeps old forecasts, so we use only what was known at 10 a.m.
CAISO day-ahead forecast	CAISO's own solar forecast.	The grid operator's own guess, a point of comparison.
EIA Form 860M	Installed solar capacity, every month.	Solar grew 24%, so 2024 and 2026 need a common scale.
CAISO curtailment reports	Solar that was turned down.	EIA measures output after the turn-down, so we need to know when it happened.

How we qualified them: public and free, documented, the same period (Mar 8, 2024 to Oct 17, 2026, 954 days), and usable at 10 a.m.

Weather data by Open-Meteo.com (CC BY 4.0).

PREPARE THE DATA

From five files to one analysis-ready table

1

Profile

Look at every file

22,896 hourly rows. 226 hours with a gap, under 1%.

2

Organize

Put it on one clock

EIA's hour labels ran 1 hour late until Dec 2, 2025. We corrected them.

3

Join

One row per hour

Five files joined on the hour. Rows counted after every join: always 22,896.

4

Enrich

Add what the model needs

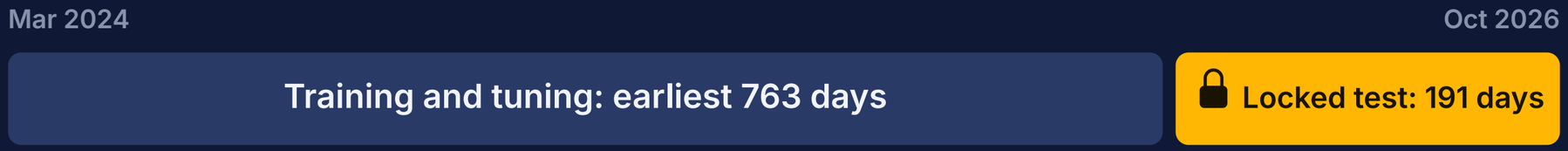
Sun up or not, season, installed capacity that month, a turned-down flag.

5

Segment

Split by time

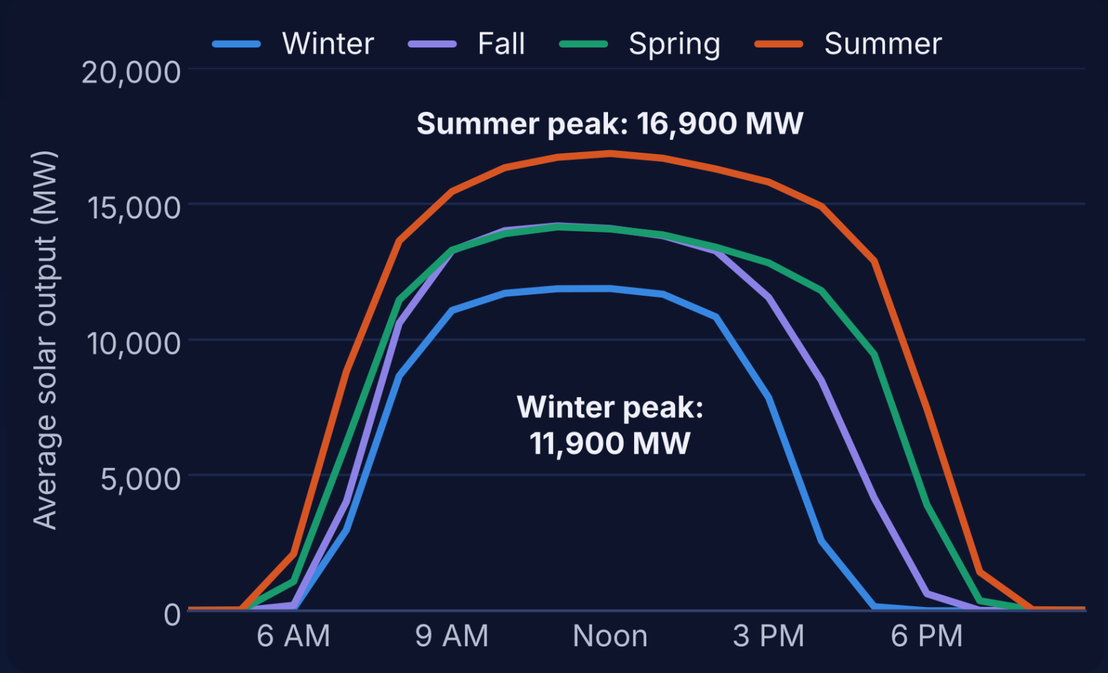
Earliest days to train and tune. The latest stay locked for the final test.



DESCRIPTIVE ANALYTICS

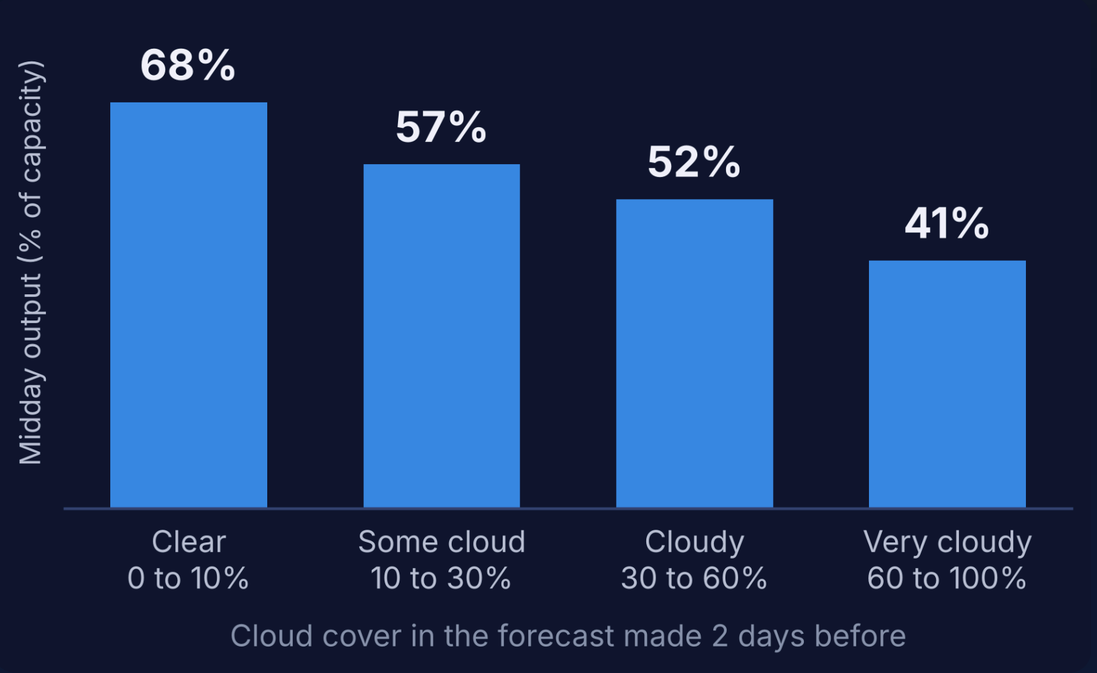
What the data shows: the sun and the clouds

A typical solar day, by season



Up at 6, peak at noon, done by 8. **Summer's peak is 42% higher than winter's.**

Cloudier forecast, less solar



Clear forecast: 68% of capacity at midday. **Very cloudy: 41%.**

Charts made in Python from our hourly table. Training and validation days (Mar 2024 to Apr 9, 2026). Midday = 10 a.m. to 1 p.m. Pacific. Sources: EIA, Open-Meteo.

DATA QUALITY

How we check quality, and what we did

WHAT WE FOUND	HOW WE FOUND IT	WHAT WE DID
Clocks off by 1 hour	Compared EIA's daily curve with CAISO's own numbers and the sun's position.	Moved EIA's labels back 1 hour before Dec 2, 2025. They now match CAISO (correlation 0.998).
Missing hours	Counted the blanks in every file.	226 of 22,896 hours, 129 from one weather outage. Flagged and left out of the scores.
Odd values at night	Checked output while the sun was below the horizon.	Small negatives set to 0 in a working copy, original kept. 351 odd hours flagged and left out of training.
Turned-down solar	Compared CAISO's curtailment reports with EIA output.	7.5% of possible solar, up to 24% in April 2026. Flagged, so we can report with and without.

Raw files are never edited. Every fix is code that anyone can re-run.

PRIVACY AND DELIVERY RISKS

Privacy is not a concern. That is the point.

PRIVACY

Open data, nothing personal

Public, grid-wide numbers and weather. No customers, no homes, no personal information.

So the whole model can be open: a base, or as is, for anyone's business case.

DELIVERY RISK 1

Free weather data has limits

Non-commercial license and a daily cap (we used about 6,500 of 10,000 calls). A business needs the paid tier.

DELIVERY RISK 2

Sources can change without notice

EIA changed its clock labels on Dec 2, 2025. Our pull script and timing check can be re-run.

DELIVERY RISK 3

Short history

2.6 years: each month appears 2 or 3 times, and our test period has no winter.

Next, in Capstone 3: build and compare forecasting models on this table.